σ_2-G1 at $V_{(3,2,1)}/n = 6$: Atom-Alphabet Decomposition of the Trace-Square Difference

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Abstract

We resolve the trace-square reframing of σ_2-G1 at the cleanest non-trivial case: $V_{(3,2,1)}$ at $n = 6$, where the standing problem of [**?**] reduces to checking that an explicit polynomial quotient lies in $\mathbb{Z}_{\geq 0}[q]$ and is palindromic. We show the σ_2-core of Π_q on $V_{(3,2,1)}/n = 6$ is exactly the degree-10 orphan palindrome $A_{10}(q) = q^{10} + 16q^9 + 105q^8 + 369q^7 + 759q^6 + 961q^5 + 759q^4 + 369q^3 + 105q^2 + 16q + 1$ already catalogued in the atom alphabet. Moreover the σ_1-core admits the clean atom-product factorisation

$$A_{(3,2,1),1}(q) \;=\; (q^2 + 4q + 1) \cdot (q^4 + 11q^3 + 21q^2 + 11q + 1) \;=\; \mathrm{atom}_E \cdot \mathrm{atom}_{\mathrm{RTL}},$$

where the second factor is OEIS A344437 row 5 (derangement right-to-left minima). The trace-square identity $\big(\mathrm{atom}_E \cdot \mathrm{atom}_{\mathrm{RTL}}\big)^2 \;=\; G_{(3,2,1),2} \;+\; 2q \cdot A_{10}$ expresses the entire σ_2-G1 statement at this case as a polynomial identity between members of the small atom alphabet.

1 Setup

Let $H_q = H_q(S_6)$ be the Iwahori–Hecke algebra over $\mathbb{Z}[q]$ with generators $T_1, \dots, T_5$ satisfying $(T_i + 1)(T_i - q) = 0$. Fix the staircase reduced word

$$w_0 \;=\; (s_1)(s_2 s_1)(s_3 s_2 s_1)(s_4 s_3 s_2 s_1)(s_5 s_4 s_3 s_2 s_1)$$

of length $\binom{6}{2} = 15$, and set $\Pi_q = \prod_i (1 + T_i)$ along this word. For each $\lambda \vdash 6$ with $V_\lambda^{H_n} \neq 0$, where $H_n = \langle s_1, s_3, \dots \rangle$, define

$$\sigma_k(\lambda, q) \;=\; e_k\big(\text{eigenvalues of } \Pi_q|_{V_\lambda}\big)$$

with e_k the k-th elementary symmetric polynomial. By [**?**, **?**],

$$\sigma_1(\lambda, q) \;=\; q^{a_\lambda}(1+q)^{b_\lambda} A_\lambda(q), \qquad \mathrm{tr}\big(\Pi_q^2|_{V_\lambda}\big) \;=\; q^{2a_\lambda}(1+q)^{2b_\lambda} F_{2,\lambda}(q),$$

with A_λ, $F_{2,\lambda}$ palindromic in $\mathbb{Z}_{\geq 0}[q]$. Newton's identity $2\sigma_2 = \sigma_1^2 - \mathrm{tr}(\Pi_q^2)$ gives, when no extra $(1+q)$-factor appears,

$$2\sigma_2(\lambda, q) \;=\; q^{2a_\lambda}(1+q)^{2b_\lambda}\big(A_\lambda^2 - F_{2,\lambda}\big).$$

The case $\lambda = (3,2,1)$ at $n = 6$ has $a_\lambda = 4$, $b_\lambda = 1$ and no extra $(1+q)$-excess: σ_2-G1 reduces to showing $(A^2 - F_2)/(2q) \in \mathbb{Z}_{\geq 0}[q]$ and palindromic, where $A = A_{(3,2,1)}$ and $F_2 = F_{2,(3,2,1)}$.

2 Computation

Working in Hoefsmit / Young seminormal form over $\mathbb{Q}(q)$ (implementation in `2026-04-29-trace-square-321/compute.py`; all matrices stored exactly as rational functions in q), we obtain

$$A(q) \;=\; q^6 + 15q^5 + 66q^4 + 106q^3 + 66q^2 + 15q + 1, \tag{1}$$

$$F_2(q) \;=\; q^{12} + 28q^{11} + 325q^{10} + 1982q^9 + 6930q^8 + 14484q^7 + 18478q^6 \tag{2}$$

$$+\, 14484q^5 + 6930q^4 + 1982q^3 + 325q^2 + 28q + 1. \tag{3}$$

A is palindromic of degree 6 with $A(1) = 270$; F_2 is palindromic of degree 12 with $F_2(1) = 66\,447$. Both are non-negative-integer-coefficient and irreducibility-checked: A factors over $\mathbb{Q}$ as

$$A(q) \;=\; \big(q^2 + 4q + 1\big)\,\big(q^4 + 11q^3 + 21q^2 + 11q + 1\big),$$

while F_2 is irreducible over $\mathbb{Q}$.

Theorem 1 (σ_2-G1 at $V_{(3,2,1)}/n = 6$, computational form)**.** *The polynomial $A^2 - F_2$ is divisible by $2q$ and the quotient*

$$A_{10}(q) \;:=\; \frac{A(q)^2 - F_2(q)}{2q}$$

satisfies $A_{10} \in \mathbb{Z}_{\geq 0}[q]$ and is palindromic. Explicitly,

$$A_{10}(q) \;=\; q^{10} + 16q^9 + 105q^8 + 369q^7 + 759q^6 + 961q^5 + 759q^4 + 369q^3 + 105q^2 + 16q + 1. \tag{4}$$

A_{10} is irreducible over $\mathbb{Q}$, has $A_{10}(1) = 3461$ (prime), and contains no cyclotomic factor.

Proof. Direct verification. Subtraction of the two explicit polynomials gives $A^2 - F_2 = 2q^{11} + 32q^{10} + 210q^9 + 738q^8 + 1518q^7 + 1922q^6 + 1518q^5 + 738q^4 + 210q^3 + 32q^2 + 2q$. The constant and degree-12 coefficients vanish, so $A^2 - F_2$ is divisible by q. Dividing by 2 (every coefficient is even) and then by q produces the expression in (**??**). Palindromy and non-negativity are immediate inspection. Irreducibility over $\mathbb{Q}$ is a finite computation (no rational roots, no quadratic or cubic divisor with palindromic shape). □

3 Atom-alphabet structure

Definition 2 (Atoms)**.** Within the atom alphabet of [**?**] we name three members appearing in the present case:

- $\mathrm{atom}_E(q) = q^2 + 4q + 1$ — Eulerian-flavoured palindrome, OEIS A008292 row 3 reversed; first σ_1-G1 atom in the program.
- $\mathrm{atom}_{\mathrm{RTL}}(q) = q^4 + 11q^3 + 21q^2 + 11q + 1$ — OEIS A344437 row 5, derangement right-to-left minima distribution.
- $\mathrm{atom}_{10}(q) = A_{10}(q)$ as in (**??**) — degree-10 orphan palindrome (no OEIS hit, irreducible over $\mathbb{Q}$, $A_{10}(1) = 3461$ prime).

Theorem 3 (σ_1-core atom factorisation at $V_{(3,2,1)}/n = 6$)**.** *The σ_1-core $A = A_{(3,2,1)}$ at $n = 6$ factors as the product of two atoms:*

$$A(q) \;=\; \mathrm{atom}_E(q) \cdot \mathrm{atom}_{\mathrm{RTL}}(q). \tag{5}$$

*Equivalently, with $A_{(3,2)} = q^4 + 9q^3 + 19q^2 + 9q + 1$ the σ_1-core of $V_{(3,2)}$ at $n = 5$ (a σ_1-G1 instance [**?**]),*

$$\mathrm{atom}_{\mathrm{RTL}}(q) \;=\; A_{(3,2)}(q) \;+\; 2q\,(q^2 + q + 1), \tag{6}$$

so $A_{(3,2,1)} = \mathrm{atom}_E \cdot A_{(3,2)} \;+\; 2q(q^2 + q + 1) \cdot \mathrm{atom}_E$ is a positive expansion in $\mathbb{Z}_{\geq 0}[q]$ of A through atoms appearing already at $n = 5$.

Proof. Direct polynomial factorisation over $\mathbb{Q}$ for (**??**). For (**??**), expand: $\mathrm{atom}_{\mathrm{RTL}} - A_{(3,2)} = (q^4 + 11q^3 + 21q^2 + 11q + 1) - (q^4 + 9q^3 + 19q^2 + 9q + 1) = 2q^3 + 2q^2 + 2q = 2q(q^2 + q + 1)$. □

Remark 4. The branching $V_{(3,2,1)} \downarrow S_5 = V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$ has σ_1-cores $A_{(3,2)} = (1, 9, 19, 9, 1)$, $A_{(3,1,1)} = \mathrm{atom}_E = q^2 + 4q + 1$, and $A_{(2,2,1)} = 1$ (rank 0 at $n = 5$). Thus the factorisation (**??**) encodes one factor of A as a σ_1-core of a branching summand, atom_E, and the other as a $2q$-perturbation of another branching summand's core, $\mathrm{atom}_{\mathrm{RTL}} = A_{(3,2)} + 2q(q^2 + q + 1)$. This is structural evidence that the Apr 24 branching reduction at $n = 5$ [**?**] generalises with a $2q$-correction, but in a multiplicative form (branching σ_1-cores enter as factors, not summands, at σ_1).

Corollary 5 (σ_2-G1 atom-alphabet identity)**.** *The σ_1-core, $\mathrm{tr}(\Pi_q^2)$-core, and σ_2-core of Π_q on $V_{(3,2,1)}$ at $n = 6$ satisfy*

$$\big(\mathrm{atom}_E \cdot \mathrm{atom}_{\mathrm{RTL}}\big)^2 \;=\; F_2(q) \;+\; 2q \cdot \mathrm{atom}_{10}. \tag{7}$$

In particular, σ_2-G1 holds at this case, with σ_2-core equal to atom_{10}.

Proof. Combine Theorems **??** and **??** and use $2\sigma_2 = \sigma_1^2 - \mathrm{tr}(\Pi_q^2)$ in core-form: $2q \cdot A_{10} = A^2 - F_2 = (\mathrm{atom}_E \cdot \mathrm{atom}_{\mathrm{RTL}})^2 - F_2$. □

4 Discussion

What this case proves and what it does not

Theorem **??** closes σ_2-G1 at $V_{(3,2,1)}$ at $n = 6$ as a polynomial identity. Together with the prior cases $V_{(5,1)}$ and $V_{(4,2)}$ (see [**?**, **?**]), all rank-≥ 2 Specht modules at $n = 6$ are now σ_2-G1.

What is *not* yet proved is a structural reason why this should be the case at general n. Identity (**??**) expresses the present case as a polynomial identity in the small atom alphabet of [**?**]. The atoms atom_E, $\mathrm{atom}_{\mathrm{RTL}}$, and atom_{10} all appear at multiple (λ, n, k) pairs in the alphabet's tabulation. The recurrence pattern of atom_{10} across three (λ, n, k) tuples — at $(6, (3, 2, 1), 2)$ (this paper), $(7, (3, 2, 1, 1), 2)$, and $(7, (3, 2, 2), 3)$ — suggests a categorical home for atom_{10} that has not yet been identified. The current status of atom_{10} is: irreducible degree-10 palindrome, no OEIS hit, no Kostka–Foulkes match for $n \leq 10$, no Gaussian binomial or parabolic Poincaré identity. It is a genuinely non-classical atom.

Atom factorisation at the σ_1-level

Theorem **??** is a new structural result at the σ_1-level. The Apr 24 σ_1-G1 proof gives positivity but not factorisation. The factorisation $A_{(3,2,1)} = \mathrm{atom}_E \cdot \mathrm{atom}_{\mathrm{RTL}}$ at $n = 6$, combined with the relation (**??**), gives a $2q$-perturbation of $A_{(3,2)}$ at $n = 5$. This should be tested at the other rank-≥ 2 shapes of S_6 — see [**?**] (in preparation).

The orphan atom and the trace-square defect

The trace-square identity (**??**) reads

$$\sigma_1^2 \;-\; F_2 \;=\; 2q \cdot \mathrm{atom}_{10}.$$

Both σ_1^2 and F_2 are sums over closed paths in the W-graph of $V_{(3,2,1)}$ — over pairs of length-N closed paths and over single closed paths of length $2N$, respectively, where $N = 15$. The difference is therefore "pairs minus doubles"; the orphan atom atom_{10} encodes a combinatorial defect of pairing. Identifying which closed-path data atom_{10} counts is the natural next concrete question.

References

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